

B.Tech – Computer Science & Engineering

Industrial Internship Report on "Rocket Delivery — Full Stack Grocery delivery Platform"

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Executive Summary

This report provides details of the Industrial Internship conducted by **Upskill Campus** in collaboration with its Industrial Partner **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship focused on developing a real-world **Full Stack Development** project assigned by UCT. The project, along with this report, was successfully completed within **6 weeks (May to June 2026)**.

My project was **Rocket Delivery**, a **Full Stack Grocery Delivery Web Application** developed using **HTML5, CSS3, JavaScript, Node.js, Express.js, MongoDB, and Mongoose**. The application enables users to register and log in securely, browse grocery products across different categories, search for products, add items to the shopping cart, apply discount coupons, place grocery orders, track their delivery status, and view previous orders. Customers can also rate delivered orders, while an integrated **Admin Dashboard** allows administrators to manage products, users, coupons, orders, and overall application statistics. The project uses **JWT Authentication** for secure login and follows the **MVC (Model View Controller)** architecture for better code organization and scalability. These technologies and features are reflected in the project structure and README.

This internship provided me with valuable hands-on experience in **frontend development, backend API development, MongoDB database management, RESTful API design, JWT authentication, CRUD operations, debugging, testing, and version control using Git and GitHub**. Developing the rocket delivery application enhanced my understanding of real-world software development practices and improved my problem-solving, analytical, and technical skills.

Overall, this internship was a great learning experience that strengthened my knowledge of full stack web development and provided practical exposure to designing and building a complete **Grocery Delivery Web Application** capable of solving real-world business requirements.

Upskill Campus & UniConverge Technologies Pvt. Ltd.

Full Stack Development Internship | May – June 2026

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1. Preface

This report summarizes the work completed during my **six-week Full Stack Development Industrial Internship** conducted by **Upskill Campus** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)**. During this internship, I designed, developed, tested, and documented **Rocket Delicery**, a full stack grocery delivery web application that provides users with an easy and convenient way to purchase groceries online.

Industrial internships play an important role in bridging the gap between academic learning and industry practices. They provide students with practical exposure to software development processes, allowing them to understand how modern web applications are planned, developed, tested, and maintained. As a B.Tech Computer Science and Engineering student, this internship helped me gain valuable hands-on experience with full stack development technologies and real-world problem solving.

My project, **Rocket Delicery**, is a grocery delivery platform that enables customers to register, log in securely, browse grocery products, search items, add products to their shopping cart, apply discount coupons, place orders, and track deliveries. The application also includes an administrative dashboard for managing users, products, customer orders, and promotional coupons. The project follows a client-server architecture and implements secure authentication using JSON Web Tokens (JWT).

Throughout the internship, Upskill Campus provided structured learning resources, weekly assignments, and technical guidance, while UniConverge Technologies Pvt. Ltd. provided the industry-oriented problem statement and development objectives. This combination helped me understand the complete software development lifecycle, from requirement analysis to deployment.

I sincerely thank the mentors and coordinators of **Upskill Campus** and **UniConverge Technologies Pvt. Ltd.** for their continuous guidance and support throughout the internship. The knowledge and practical experience gained during this program have strengthened my technical skills and prepared me for future opportunities in software development.

2. Introduction

2.1 About UniConverge Technologies Pvt. Ltd.

UniConverge Technologies Pvt. Ltd. (UCT) is a technology-driven company established in **2013** with a focus on providing innovative digital transformation solutions for industries. The organization specializes in developing software and hardware solutions that help businesses improve operational efficiency, productivity, and Return on Investment (ROI).

UCT works in various domains such as **Internet of Things (IoT), Cloud Computing, Cyber Security, Artificial Intelligence, Machine Learning, Java Full Stack Development, Python Development, Frontend Development, and Industrial Automation**. The company provides industrial training and internship programs that enable students to gain practical knowledge through real-world software development projects.

Some of the major technology areas supported by UCT include:

- Internet of Things (IoT) Solutions
- Cloud Computing (AWS & Microsoft Azure)
- Java and Python Full Stack Development
- Machine Learning Applications
- Industrial Automation
- Smart Factory Solutions
- LoRaWAN-based Smart City Applications

Through collaboration with educational institutions and training partners, UCT provides students with opportunities to work on industry-oriented projects that improve both technical and professional skills.

2.2 About Upskill Campus (USC)

Upskill Campus (USC) is an online learning platform that provides students with industry-oriented training, certification programs, internships, and career development opportunities. It collaborates with organizations such as UniConverge Technologies Pvt. Ltd. to bridge the gap between academic education and industry requirements.

The platform offers structured learning through recorded sessions, live mentorship, weekly assignments, quizzes, and project-based learning. Students gain practical exposure by working on real-world applications while receiving continuous evaluation and guidance from industry professionals.

The primary objective of Upskill Campus is to help students build technical skills, improve employability, and prepare them for successful careers in the software industry.

Website: <https://www.upskillcampus.com/>

2.3 Objectives of this Internship Program

The objectives of this internship program were:

- To understand the software development lifecycle by developing a real-world application.
- To strengthen frontend and backend development skills using modern web technologies.
- To gain practical knowledge of REST API development and database management.
- To implement secure user authentication using JWT.
- To improve problem-solving and debugging skills through project development.
- To understand project organization using the MVC architecture.
- To enhance documentation, testing, and version control skills using Git and GitHub.
- To develop a production-ready full stack web application.

2.4 References

The following resources were referred to during the development of the project:

1. Node.js Official Documentation – <https://nodejs.org/docs>
2. Express.js Official Documentation – <https://expressjs.com/>
3. MongoDB Documentation – <https://www.mongodb.com/docs/>
4. Mongoose Documentation – <https://mongoosejs.com/>
5. MDN Web Docs – <https://developer.mozilla.org/>
6. JWT Documentation – <https://jwt.io/>
7. GitHub Repository – <https://github.com/dhanaranjan-dev/UPSKILLS.git>

2.5 Glossary

Term	Definition
HTML5	HyperText Markup Language version 5 — webpage structure
CSS3	Cascading Style Sheets version 3 — webpage styling
JavaScript	Programming language used for developing interactive web applications.
Node.js	JavaScript runtime environment used for server-side development.
Express.js	Lightweight web framework used for building REST APIs.
MongoDB	NoSQL database used to store application data.
MVC	Model-View-Controller architecture used for organizing application components.
REST API	Interface that enables communication between the frontend and backend using HTTP requests.

3. Problem Statement

The rapid growth of e-commerce has significantly transformed the way people purchase daily essentials. Grocery delivery applications have become increasingly popular because they provide customers with the convenience of ordering groceries from their homes while saving time and effort. However, developing a complete grocery delivery platform requires knowledge of frontend development, backend services, database management, authentication, and secure communication between different system components.

The problem statement assigned during this internship was to design and develop a **Full Stack Grocery Delivery Web Application** that provides a seamless shopping experience for customers while allowing administrators to efficiently manage products, orders, and users.

The application was expected to address the following real-world requirements:

- Users should be able to register and log in securely.
- Customers should be able to browse grocery products from different categories.
- Users should be able to search products quickly using keywords.
- Customers should be able to add, update, and remove items from their shopping cart.
- Discount coupons should be supported during checkout.
- Customers should be able to place grocery orders with complete delivery details.
- Users should be able to track the status of their orders after placing them.
- Customers should be able to view previous orders and submit ratings after successful delivery.
- Administrators should be able to manage products, users, coupons, and customer orders through a centralized dashboard.
- The application should provide a responsive interface that works efficiently on desktop and mobile devices.

The major challenge of this project was to integrate all these functionalities into a secure, scalable, and user-friendly full stack application using open-source technologies within the internship duration.

4. Existing and Proposed Solution

4.1 Existing Solutions and Limitations

Several online grocery delivery platforms such as **Blinkit, BigBasket, Zepto, and Instamart** provide fast grocery delivery services. These platforms offer a wide range of products, digital payment options, and real-time delivery tracking. However, from a student learning perspective, they have several limitations:

- Their source code is proprietary and cannot be accessed for learning purposes.
- They use highly complex architectures involving multiple microservices and cloud infrastructure.
- Their systems require expensive third-party services such as payment gateways, cloud hosting, and mapping services.
- They are designed for large-scale commercial environments, making them difficult to understand and implement as an academic project.
- Developing a similar production-level application requires significant infrastructure and financial resources.

Therefore, building a simplified but feature-rich grocery delivery application provides an excellent opportunity to understand full stack web development concepts while implementing real-world functionality.

4.2 Proposed Solution — Rocket delivery

Rocket Delivery is a full stack grocery delivery web application developed using modern web technologies to provide a smooth and efficient online grocery shopping experience.

The application enables customers to create accounts, log in securely, browse grocery products by category, search products, add items to the shopping cart, apply discount coupons, place grocery orders, and monitor order status through an intuitive interface. After successful delivery, users can also provide ratings and reviews based on their shopping experience.

The backend of the application is developed using **Node.js** and **Express.js**, while **MongoDB** serves as the database for storing user accounts, products, orders, and coupon information. Secure authentication is implemented using **JSON Web Tokens (JWT)** to protect user data and restrict unauthorized access.

An **Admin Dashboard** is provided to simplify management tasks. Administrators can:

- View all registered users.
- Manage customer orders.
- Update order status.
- Create and manage discount coupons.
- Monitor overall application statistics.

The application follows the **Model-View-Controller (MVC)** architecture, which improves code organization, maintainability, and scalability. RESTful APIs are used for communication between the frontend and backend, ensuring efficient data exchange and a modular system design.

The proposed solution successfully demonstrates the implementation of a real-world grocery delivery

platform using open-source technologies while providing practical exposure to modern full stack web development practices.

GitHub Repository

<https://github.com/dhanaranjan-dev/UPSKILLS.git>

5. Proposed Design / Model

Rocket Delivery follows a **Client-Server Architecture**, where the frontend communicates with the backend using REST APIs. The backend processes client requests, performs authentication, interacts with the MongoDB database, and returns appropriate responses. This architecture provides better scalability, maintainability, and separation of concerns.

5.1 High Level Diagram

System Architecture Overview:



5.2 Low Level Diagram — Backend File Structure

File / Folder	Purpose
server.js	Starts the Express server and configures middleware.
config/db.js	Establishes MongoDB database connection.
models/User.js	Stores user information and authentication details.
models/Product.js	Defines grocery product schema.
models/Order.js	Stores customer order information.
models/Coupon.js	Maintains coupon codes and discount details.
routes/auth.js	Handles user registration and login APIs.
routes/products.js	Manages product-related APIs.
routes/orders.js	Handles order placement and tracking.
routes/coupons.js	Validates and applies coupons.
routes/admin.js	Provides administrative APIs.
middleware/auth.js	Verifies JWT tokens and secures protected routes.
seed.js	Inserts demo products, users, and coupons into the database.

5.3 Interfaces (Application Pages)

Home Page

The Home Page displays featured grocery products, popular categories, offers, and navigation options. Users can quickly browse products and explore different grocery sections.

Product Page

Displays all available grocery products with category filters and search functionality. Customers can easily locate products based on their requirements.

Product Details

Shows complete information about the selected product, including description, price, category, stock availability, and the option to add the product to the shopping cart.

Shopping Cart

Allows users to view selected products, update quantities, remove items, apply discount coupons, and view the total order amount before checkout.

Checkout Page

Collects delivery address details and confirms the customer's grocery order. The system validates coupons and generates an order successfully.

Order Tracking

Displays the current delivery status of placed orders. Customers can monitor the progress from order confirmation to successful delivery.

Order History

Provides a complete list of previously placed orders along with their details and current status. Delivered orders can also be rated by customers.

User Profile

Allows customers to manage their personal information and view their account details.

Admin Dashboard

The Admin Dashboard enables administrators to:

- View all registered users.
- Manage customer orders.
- Update delivery status.
- Create and manage discount coupons.
- Monitor application statistics and revenue.

6. Performance Test

Performance testing was conducted to verify that the **Rocket Delivey Grocery Delivery Application** performs efficiently under normal usage conditions. The application was tested on different modules to ensure that all major functionalities work correctly and provide a smooth user experience.

6.1 Test Plan / Test Cases

S.No	Test Case	Expected Result	Outcome
1	Register a new user	User account created successfully	PASS
2	Login with valid credentials	User redirected to Home page	PASS
3	Invalid login attempt	Error message displayed	PASS
4	Search for a grocery product	Matching products displayed instantly	PASS
5	Add product to cart	Product added and cart count updated	PASS
6	Update product quantity	Cart total updated correctly	PASS
7	Apply coupon code	Discount applied successfully	PASS
8	Place grocery order	Order created successfully with Order ID	PASS
9	Track placed order	Order status updated correctly	PASS
10	Admin login	Dashboard displayed successfully	PASS
11	Update order status	Customer order updated successfully	PASS
12	Mobile responsive layout	Application displayed correctly on different screen sizes	PASS

6.2 Test Procedure

The application was tested using different scenarios to verify both frontend and backend functionalities.

- Google Chrome was used for functional testing.
- Backend APIs were tested using the Node.js development server.
- MongoDB was used to verify data storage and retrieval.
- JWT authentication was tested using both valid and invalid login credentials.
- Shopping cart functionality was tested by adding, updating, and removing products.
- Coupon validation was tested using valid and invalid coupon codes.
- Order placement and tracking were tested using different customer accounts.
- Admin functionalities such as managing users, coupons, and orders were verified.
- Responsive design was tested using Chrome Developer Tools on mobile, tablet, and desktop screen sizes.

6.3 Performance Outcome

The Rocket Delivery application performed successfully during all testing phases. The overall performance observations are listed below:

- Fast page loading with responsive user interface.
- Smooth navigation between different application modules.
- Secure authentication using JWT.
- Product search returns results quickly.
- Shopping cart updates instantly without page refresh.
- Coupon validation works accurately.
- Orders are created and stored successfully in MongoDB.
- Order tracking displays updated delivery status.
- Admin Dashboard efficiently manages users, coupons, and customer orders.
- The application performs well on desktop, tablet, and mobile devices.

The testing results confirmed that the developed Grocery Delivery Application satisfies the

7. My Learnings

The Rocket Delivery project provided valuable practical experience in developing a complete Full Stack Web Application. Throughout the internship, I gained technical knowledge as well as professional skills that strengthened my understanding of software development.

Technical Skills Gained

- Developed a complete Full Stack Grocery Delivery Application.
- Improved frontend development using HTML5, CSS3, and JavaScript.
- Built RESTful APIs using Node.js and Express.js.
- Learned MongoDB database design and management using Mongoose.
- Implemented secure user authentication using JWT.
- Understood the MVC architecture for organizing application code.
- Developed CRUD operations for products, users, coupons, and orders.
- Implemented shopping cart and order management functionalities.
- Improved debugging and error handling skills.
- Learned version control using Git and GitHub.

Professional Skills Gained

- Improved problem-solving and analytical thinking.
- Learned project planning and task management.
- Enhanced documentation and technical writing skills.
- Improved communication through weekly progress reporting.
- Understood the importance of clean code and software testing.
- Developed confidence in building industry-level web applications.

This internship helped me understand how real-world web applications are designed, developed, tested, and maintained. It enhanced both my technical abilities and professional confidence, preparing me for future software development opportunities.

8. Future Work Scope

The Rocket Delivery Grocery Delivery Application successfully implements the core functionalities required for an online grocery delivery system. However, due to the limited internship duration, several advanced features were not included in the current version. These enhancements can be implemented in future versions to improve the application's scalability, security, and user experience.

Phase 2 – Backend Enhancements

The following backend improvements can be implemented in future releases:

- Integrate secure online payment gateways such as **Razorpay**, **Stripe**, or **PayPal**.
- Implement real-time order tracking using **Socket.IO**.
- Send order confirmation and delivery notifications through email services.
- Add SMS notifications for order status updates.
- Improve API security using rate limiting and request validation.
- Optimize database queries for better performance.
- Implement inventory management with automatic stock updates.
- Add role-based access control for administrators and delivery partners.

Phase 3 – Frontend Enhancements

The user interface can be enhanced by introducing additional modern features, including:

- Migrate the frontend to **React.js** for improved performance and component reusability.
- Implement Progressive Web App (PWA) support for mobile installation.
- Add Google Maps integration for accurate delivery location tracking.
- Introduce Wishlist functionality for saving favorite products.
- Improve product recommendation using customer purchase history.
- Enhance product filtering with advanced search options.
- Provide dark mode support for improved accessibility.
- Improve UI animations and overall user experience.

Phase 4 – Business Features

To make Rocket Delivery more suitable for commercial deployment, the following business-oriented features can be introduced:

- Multi-vendor support allowing multiple grocery stores to sell products.
- Delivery partner management system.
- Real-time delivery assignment.
- Customer loyalty and reward points system.
- Subscription plans for premium users.
- AI-based product recommendations.
- Sales analytics and reporting dashboard.
- Customer support chat system.
- Invoice generation and GST billing.
- Multi-language support for wider accessibility.

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